

A Comparative Analysis of NLP and Machine Learning Approaches for Phishing Email Detection

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Abstract—Phishing attacks remain one of the most prevalent cybersecurity threats, with attackers continuously adapting their techniques to bypass traditional detection methods. This study developed an AI-powered phishing email detection system using natural language processing (NLP) and machine learning. Using the CEAS-08 email dataset comprising 39,154 labeled samples, both traditional machine learning models (Logistic Regression and Random Forest) and a BERT-based deep learning model were trained and evaluated. Text features were extracted using TF-IDF vectorization for the traditional models. All three models achieved 99% accuracy. BERT produced the fewest total misclassifications (50) with the highest phishing precision (1.00), while Logistic Regression achieved the highest phishing recall (1.00). Despite BERT's marginal advantage, traditional ML models delivered comparable results with significantly less training time and computational cost.

Keywords—phishing detection, machine learning, BERT, NLP, TF-IDF, email security, deep learning

I. INTRODUCTION

Cybersecurity is critical as organizations and individuals depend on digital communication. Email remains the primary attack vector for cybercriminals, with the Anti-Phishing Working Group (APWG) recording over 1.28 million phishing attacks in Q2 2023 [1]. Approximately 82 percent of malware infections begin with a phishing email [1].

Phishing emails exploit human psychology using urgency, authority, and familiar business language to trick victims [2][3]. Traditional detection methods like blacklists, signature matching, and rule-based filters are reactive and cannot keep up with evolving attacks [5]. There is a need for intelligent, adaptive systems that can learn to detect phishing automatically.

This study addresses the lack of direct, controlled comparisons between traditional machine learning (ML) and transformer-based deep learning approaches for phishing detection. Three models—Logistic Regression, Random Forest, and BERT—are trained and evaluated on the same CEAS-08 dataset using identical

preprocessing and evaluation metrics to provide a fair comparison.

The research questions guiding this study are: (RQ1) How effectively can traditional ML models detect phishing emails using TF-IDF features? (RQ2) Does BERT significantly outperform traditional ML on the same dataset? (RQ3) What tradeoffs exist between accuracy, training time, and computational cost?

II. RELATED WORK

A. The Phishing Threat

Phishing is a continuously growing threat. It exploits human psychology as a form of social engineering, with a person's experience and personality traits influencing their vulnerability [2]. Wright et al. examined influence techniques in phishing attacks and found that urgency, authority, and social proof vary in effectiveness depending on victim awareness [3]. Expel's analysis of over 10,000 malicious emails found that top phishing keywords were business terms like "required," "invoice," and "document" [4].

B. Traditional Detection Methods

Prior phishing defenses relied on blacklisting, whitelisting, signature-based detection, and rule-based filtering. Salloum et al. found these approaches cannot effectively prevent sophisticated or zero-day attacks [5]. Kyaw et al. confirmed that research has shifted toward ML and deep learning specifically because traditional methods fall short [6].

C. Machine Learning Approaches

Researchers have applied Logistic Regression, Random Forest, SVM, and Naive Bayes to phishing classification with promising results. Altwaijry et al. confirmed that ML-based approaches outperform rule-based methods [7]. Traditional ML models remain valuable due to their computational efficiency and interpretability, though they depend on manually engineered features like TF-IDF vectors.

D. Deep Learning and Transformers

Transformer-based models have achieved detection accuracy over 98% in multiple studies. Splunk fine-tuned BERT on over 181,000 labeled emails and achieved an F1 score of 0.99 [8]. Songailaitė et al. fine-tuned DistilBERT, TinyBERT, and RoBERTa, all achieving F1 scores exceeding 0.985 [9]. BERT understands contextual meaning rather than just counting words, making it more effective against sophisticated phishing. However, it requires significantly more computing power.

E. Research Gap

Existing studies typically test only traditional ML or only deep learning. Few conduct direct comparisons on the same dataset with identical metrics. This study addresses this gap by evaluating Logistic Regression, Random Forest, and BERT on the CEAS-08 dataset under identical conditions.

III. METHODOLOGY

A. Dataset

The CEAS-08 dataset contains 39,154 labeled emails: 21,842 phishing and 17,312 legitimate. It includes sender, receiver, date, subject, body, label, and URL count fields. The dataset was split 80/20 into training (31,323) and testing (7,831) sets.

B. Preprocessing

Preprocessing included removing null values, converting labels to binary, lowercasing text, removing HTML tags, and stripping special characters. Exploratory analysis revealed phishing emails averaged 801 characters versus 2,542 for legitimate emails. Urgency keywords and URL counts were not strong differentiators.

C. Feature Extraction

For traditional ML models, TF-IDF vectorization converted email text into numerical vectors with a maximum of 5,000 features and English stopword removal. For BERT, text was tokenized using the bert-base-uncased tokenizer with a maximum length of 256 tokens.

D. Models

Logistic Regression: A linear classifier trained on TF-IDF features, optimized with max_iter=1000. **Random Forest:** An ensemble of 100 decision trees trained on TF-IDF features. **BERT:** The bert-base-uncased model fine-tuned for 3 epochs with a learning rate of 2e-5, batch size of 16, and AdamW optimizer on GPU.

E. Evaluation

All models were evaluated using precision, recall, F1-score, and confusion matrices on the same test set of 7,831 emails.

IV. RESULTS

A. Overall Performance

All three models achieved 99% accuracy on the test set. Table I presents the detailed comparison.

TABLE I: MODEL PERFORMANCE COMPARISON

Model	Prec.	Rec.	F1	Errors
Log. Reg.	0.99	1.00	0.99	63
Ran. For.	0.99	0.99	0.99	55
BERT	1.00	0.99	0.99	50

B. Error Analysis

The models differed in their error patterns. Logistic Regression produced 42 false positives and 21 false negatives (63 total errors). Random Forest produced 31 false positives and 24 false negatives (55 total). BERT produced only 11 false positives but 39 false negatives (50 total). BERT achieved the best precision for phishing detection (1.00), while Logistic Regression achieved the best recall (1.00).

TABLE II: CONFUSION MATRIX SUMMARY

Model	TN	TP	FP	FN
Log. Reg.	3,448	4,320	42	21
Ran. For.	3,459	4,317	31	24
BERT	3,479	4,302	11	39

C. Training Efficiency

Logistic Regression trained in approximately 5 seconds, Random Forest in 2 minutes, and BERT required approximately 30 minutes on a T4 GPU via Google Colab. This represents a 360x difference in training time between the fastest and slowest models for a marginal difference of 13 fewer errors.

V. DISCUSSION

All three models achieved 99% accuracy, demonstrating that both traditional ML and deep learning can effectively detect phishing emails in the CEAS-08 dataset. BERT minimizes false alarms (11 FP) while Logistic Regression minimizes missed threats (21 FN), presenting a clear tradeoff depending on deployment priorities.

The null hypothesis (H0) is partially supported: while BERT achieved the fewest total errors and highest

precision, the overall performance difference was marginal. For this dataset with clear phishing patterns, traditional ML models deliver comparable results with significantly less computational cost.

BERT's advantage would likely increase on more sophisticated, modern phishing datasets where contextual understanding becomes more critical. The CEAS-08 dataset's spam-style phishing is relatively straightforward for TF-IDF-based models to distinguish.

For resource-constrained organizations, Logistic Regression or Random Forest offer an effective and practical solution. BERT becomes more justified when minimizing false positives is critical or when phishing language closely mimics legitimate communication.

VI. CONCLUSION

This study compared Logistic Regression, Random Forest, and BERT for phishing email detection on the CEAS-08 dataset. All three achieved 99% accuracy. BERT had the fewest total errors (50) and best precision, while Logistic Regression had the best recall. Traditional ML models delivered comparable performance with a fraction of the training time and cost.

Future work should test on modern phishing datasets, explore lighter alternatives such as DistilBERT, evaluate against AI-generated phishing emails, and integrate detection models into live email filtering systems. Expanding features to include header analysis, URL inspection, and attachment scanning could further improve detection capabilities.

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